

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12385247
Kind Code	B2
Date of Patent	August 12, 2025
Inventor(s)	Brekke; Steven

Girder tie

Abstract

A girder tie for connecting a building component to a rigid rod to resist uplift forces applied to the building component comprises a connector including a building component connector and a rigid rod connector coupled to the building component connector. The rigid rod connector attaches to the rigid rod. The building component connector includes first and second back flanges free of direct connection to one another. The first and second back flanges can each be attached to the building component. A washer is disposed between the rigid rod connector and a nut on the rigid rod that secures the rigid rod to the girder tie. The washer includes at least one back flange brace to inhibit the first and second back flanges from moving relative to one another when the building component experiences the uplift forces.

Inventors:	Brekke; Steven (Lakeville, MN)
Applicant:	Columbia Insurance Company (Omaha, NE)
Family ID:	76439577
Assignee:	Columbia Insurance Company (Omaha, NE)
Appl. No.:	18/469341
Filed:	September 18, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20240003131 A1	Jan. 04, 2024

Related U.S. Application Data

division parent-doc US 17128264 20201221 US 11821199 child-doc US 18469341
us-provisional-application US 62950455 20191219

Publication Classification

Int. Cl.: E04B1/38 (20060101); E04H9/14 (20060101)

U.S. Cl.:

CPC E04B1/388 (20230801); E04H9/14 (20130101); E04B2001/389 (20230801)

Field of Classification Search

CPC: E04B (1/388); E04B (2001/389); E04H (9/14)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
4183305	12/1979	Payne	432/250	C21D 9/70
4665672	12/1986	Commins	52/702	F16B 9/058
6112495	12/1999	Gregg et al.	N/A	N/A
6513290	12/2002	Leek	52/297	E04B 1/2604
6550200	12/2002	Mueller	N/A	N/A
6715258	12/2003	Mueller	N/A	N/A
7707785	12/2009	Lin	N/A	N/A
7971411	12/2010	Commins	52/293.3	E04B 1/2604
8234826	12/2011	Proffitt, Jr.	N/A	N/A
8387321	12/2012	diGirolamo	52/713	E04B 2/58
8555580	12/2012	Vilasineekul	52/696	E04B 1/2604
8769887	12/2013	Proffitt, Jr.	N/A	N/A
8844229	12/2013	Hohmann, Jr.	N/A	N/A
9297161	12/2015	Zimmerman et al.	N/A	N/A
10179992	12/2018	Brekke	N/A	N/A
10415614	12/2018	Belinda	N/A	F16B 5/02
10480177	12/2018	Siddhartha et al.	N/A	N/A
2002/0066247	12/2001	Leek	52/297	E04B 1/2604
2003/0213194	12/2002	Kawai	52/261	E04C 3/07
2006/0213136	12/2005	Lin	N/A	N/A
2008/0098669	12/2007	Lin	52/92.2	E04B 7/045
2009/0107082	12/2008	Commins	52/745.12	E04H 9/14
2009/0113839	12/2008	Carr	N/A	N/A
2015/0184370	12/2014	Brekke et al.	N/A	N/A
2015/0218832	12/2014	Peters et al.	N/A	N/A

OTHER PUBLICATIONS

Simpson Strong-Tie Company, Inc., Simpson Strong-Tie Wood Construction Connectors VGT Catalog Page (003), Girder Tiedowns, pp. 278-279, 2019. cited by applicant

Primary Examiner: Katcheves; Basil S

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a divisional of U.S. patent application Ser. No. 17/128,264 filed Dec. 21, 2020, titled Girder Tie, which application claims priority to U.S. Provisional Application No. 62/950,455, filed Dec. 19, 2019, the entireties of which are incorporated by reference.

FIELD

(1) The present disclosure generally relates to girder ties used to resist uplift loads in buildings and other structures.

BACKGROUND

(2) Girder ties are used to resist uplift loads of building components, such as joists, beams, trusses, etc. Girder ties are commonly used in buildings located in high wind areas (e.g., hurricane or tornado areas) to resist the uplift forces applied to building components by winds blowing into, over, and/or around the building. One conventional type of girder tie connects the building component to a rigid rod that is anchored to a part of the building such as a foundation or a wall. When an uplift force is applied to the building component, the connection between the rigid rod and the girder tie resists the uplift force, holding the building component in position.

SUMMARY

(3) In one aspect, a girder tie for connecting a building component to a rigid rod to resist uplift forces applied to the building component comprises a connector. The connector includes a building component connector and a rigid rod connector coupled to the building component connector. The rigid rod connector is configured to attach to the rigid rod. The building component connector includes first and second back flanges free of direct connection to one another. The first and second back flanges are each configured to attach to the building component. A washer is configured to be disposed between the rigid rod connector and a nut on the rigid rod that secures the rigid rod to the girder tie. The washer includes at least one back flange brace configured to inhibit the first and second back flanges from moving relative to one another when the building component experiences the uplift forces.

(4) In another aspect, a girder tie for connecting a building component to a rigid rod to resist uplift forces applied to the building component comprises a connector. The connector includes a building component connector configured to attach to the building component and a rigid rod connector coupled to the building component connector. The rigid rod connector is configured to attach to the rigid rod. The rigid rod connector is configured to form a moment couple with the rigid rod to resist the uplift forces applied to the building component when the building component experiences the uplift forces. A nut is configured to be threaded onto the rigid rod to secure the rigid rod to the girder tie.

(5) Other objects and features of the present disclosure will be in part apparent and in part pointed out hereinafter.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a perspective of a fragmentary portion of a building showing a girder tiedown assembly according to one embodiment of the present disclosure tying a roof truss of the building to a column or stud(s) of the building;

- (2) FIG. 2 is a perspective of a girder tie of the girder tiedown assembly;
 - (3) FIG. 3 is a perspective of a connector of the girder tie;
 - (4) FIG. 4 is a back side perspective of a washer of the girder tie;
 - (5) FIG. 5 is a top view of a connector blank for forming the connector;
 - (6) FIG. 6 is a top view of a washer blank for forming the washer;
 - (7) FIG. 7 is a front perspective of a fragmentary portion of a building showing a first girder tiedown assembly according to another embodiment of the present disclosure tying a truss of the building to a column or stud(s) of the building;
 - (8) FIG. 8 is a rear perspective the fragmentary portion of the building shown in FIG. 7 showing a second girder tiedown assembly according to another embodiment of the present disclosure tying the truss to the column or stud(s); and
 - (9) FIG. 9 is a perspective of a girder tie of the second girder tiedown assembly.
- (10) Corresponding reference characters indicated corresponding parts throughout the drawings.

DETAILED DESCRIPTION OF THE DISCLOSURE

(11) Referring to FIG. 1, one embodiment of a girder tiedown assembly constructed according to the teachings of the present disclosure is indicated generally at reference numeral **10**. As shown in FIG. 1, the girder tiedown assembly **10** is used to tie or anchor one building component to a supporting member in order to resist any uplift forces that are applied to the building component. In the illustrated embodiment, the girder tiedown assembly **10** is used to tie a roof truss T (e.g., building component) to a stud(s) C of a wall (e.g., supporting member) of the building to counteract any uplift forces that may lift the roof truss generally upward and away from the stud(s). However, it is understood that the girder tiedown assembly **10** can be used to tie generally any type of building component down such as a joist, a beam, another type of truss, a column, etc. It is also understood that the girder tiedown assembly **10** can be used to tie a building component T to support members besides studs C such as columns and concrete walls (e.g., foundation walls).

(12) The girder tiedown assembly **10** includes a holdown or anchor **12**, a rigid rod **14** and a girder tie **100**. The holdown **12** is secured to the stud C and the girder tie **100** is secured to the truss T (broadly, building component), with the rigid rod **14** interconnecting the holdown and girder tie to prevent the truss from being lifted relative to the wall. One example of a suitable holdown is the PHD/DTB Holdowns available from MiTek USA, Inc., St. Louis, Missouri. Nuts **16** are used to secure the rigid rod to the holdown **12** and girder tie **100**—i.e., the rigid rod is at least partially threaded. The girder tie **100** connects the truss T to the rigid rod **14** to resist the uplift forces applied to the truss. In the illustrated embodiment, the girder tiedown assembly **10** is used with wood frame construction with the rigid rod **14** extending through a top plate B of the wall to interconnect the holdown **12** and the girder tie **100**. The girder tiedown assembly **10** can be used with other types of construction. For example, the girder tiedown assembly **10** can be used to tie the truss T a concrete wall (e.g., supporting member). In that embodiment, the girder tiedown assembly **10** may not include the holdown **12**. Instead, the rigid rod **14** can be embedded in the concrete wall (not shown).

(13) Referring to FIGS. 2-4, the girder tie **100** includes a connector **102** and a washer **150**. The connector **102** is configured to be attached to the truss T and the rigid rod **14**. The connector **102** includes a building component connector **104** and a rigid rod connector **106**. The building component connector **104** and rigid rod connector **106** are coupled to one another and, more preferably, fixed to one another. The building component connector **104** is configured to attach to the truss T. The building component connector **104** includes first and second back flanges **108** (broadly, at least two back flanges). The first and second back flanges **108** are each configured to attach to the truss T. In one embodiment, the back flanges **108** are sized and shaped to attach to a 2×6 or larger piece of dimensioned lumber. In the illustrated embodiment, each back flange **108** includes a plurality of fastener holes **110** sized and shaped to receive fasteners **18** (FIG. 1), such as screws, bolts, nails, etc., to secure the back flanges and the building component connector **104** to

the truss T. The first and second back flanges **108** are free of direct connection to one another. In one embodiment, the first and second back flanges **108** are spaced apart by about 1/16 inch (1.6 mm). However, the first and second back flanges **108** could be touching one another within the scope of the present invention. The first and second back flanges **108** are generally planar, with planar rear surfaces that engage the truss T when the building component connector **104** is secured to the truss T. The back flanges **108** can have generally any shape. Each back flange **108** may also include a notch or opening **118** configured to receive a portion of the washer **150**, as described in more detail below. In the illustrated embodiment, each back flange **108** includes a notch **118** extending from a lower edge margin of the back flange.

(14) The orientation of the girder tie **100** in FIG. 2 provides the point of reference for the terms defining relative locations and positions of structures and components of the girder tie, including but not limited to the terms “upper,” “lower,” “left,” “right,” “back,” “rear,” “front,” as used throughout the present disclosure.

(15) The connector **102** includes a flange or rib **112** extending forward from each back flange **108**. In the illustrated embodiment, each rib **112** extends from an inner edge margin (e.g., the edge margin closest to the other back flange) of a corresponding back flange **108**. Each rib **112** interconnects the back flanges **108** and the rigid rod connector **106**. Accordingly, the ribs **112** extend from the back flange **108** to the rigid rod connector **106**. The ribs **112** are generally vertical, when the connector **102** is secured to the truss T. The ribs **112** are generally perpendicular to the back flanges **108**. In the illustrated embodiment, the ribs **112** generally extend along the entire height of the back flanges **108**. The ribs **112** are adjacent to and generally parallel to one another for reasons that will become apparent. In one embodiment, a distance between the ribs **112** is less than or equal to 1/16 inch (1.6 mm). As described in more detail below, the ribs **112** facilitate the bracing of first and second back flanges **108** to inhibit the movement of the first and second back flanges relative to one another when the truss T experiences the uplift forces. The ribs **112** also generally stiffen and strengthen the back flanges **108**.

(16) The rigid rod connector **106** is configured to attach to the rigid rod **14**. The rigid rod connector **106** defines a central passage **114** sized and shaped to receive the rigid rod **14**. The rigid rod connector **106** includes a generally cylindrical wall or tube **116** that defines the central passage **114**. The ribs **112** extend from the cylindrical wall **116**. In the illustrated embodiment, the ribs **112** extend from opposite side edge margins of the generally cylindrical wall **116**. Accordingly, in the illustrated embodiment, the cylindrical wall **116** is circumferentially discontinuous. As will become apparent, this discontinuity in the cylindrical wall **116** allows the connector **102** to be stamped from a single sheet of material, as described in more detail below. The rigid rod connector **106** has a height extending from a lower end to an upper end. In one embodiment, the height of the rigid rod connector **106** is about 2 $\frac{3}{8}$ inches (6 cm). In one embodiment, the height of the rigid rod connector **106** is about half of the height of the back flanges **108**. Other configurations of the rigid rod connector **106** are within the scope of the present disclosure.

(17) The rigid rod connector **106** is configured to form a moment couple with the rigid rod **14** to resist the uplift forces applied to the truss T when the truss experiences the uplift forces. The rigid rod connector **106** is configured to engage the rigid rod **14** at a minimum of least two longitudinally spaced apart locations on the rigid rod when the truss T experiences the uplift forces to form the moment coupled with the rigid rod. Specifically, upper and lower ends of the rigid rod connector **106** engage the rigid rod **14** to form the moment couple as described in more detail below.

(18) By forming a moment couple with the rigid rod **14**, the girder tie **100** is able to resist larger uplift forces than conventional girder ties. Conventional girder ties do not form a moment couple with the rigid rod **14** because conventional girder ties engage the rigid rod at only one longitudinal location. In fact, some conventional girder ties permit the girder tie and rigid rod to pivot relative to one another, which completely prevents any moment couple from forming.

(19) When the girder tie **100** is subjected to loads (e.g., uplift forces), the failure mode for the

girder tie is being pulled from the truss T. Specifically, the fasteners **18** securing the girder tie **100** to the truss T are pulled out from (e.g., withdraw from) the truss when a sufficient amount of force is applied. When subject to uplift loads capable of causing failure, the nut **16** securing the rod **14** to the girder tie **100** is, in effect, driven down against the top of the cylindrical wall **116** of the rigid rod connector **106**. The force applied to the rigid rod connector **106** is spaced from the back flanges **108** and therefore urges the girder tie **100** generally to pivot or rotate about its lowest most point (or thereabout) that engages the truss T. This movement tends to pry the fasteners **18** out from the truss T. The fasteners **18** resist this withdrawal movement, and the girder tie **100** is constructed to provide substantial additional resistance to pivoting and withdrawal. As the girder tie **100** begins to bend and pivot, the rigid rod connector **106** engages the rigid rod **14** at generally two spaced apart locations, one generally at the upper end of the rigid rod connector and another at the lower end of the rigid rod connector. This forms the moment couple between the rigid rod connector **106** and the rigid rod **14**. Because of the moment couple, in order for the girder tie **100** to continue to pivot and move away from the truss T (e.g., in order for the girder tie to completely fail), the girder tie must bend the rigid rod **14**. Accordingly, the resistance to bending provided by employing the stiffness of the rigid rod **14** increases the amount of the uplift force the girder tie **100** can support over conventional girder ties by reducing the withdrawal forces applied to the fasteners **18**.

(20) The loads applied during uplift can also have a tendency to separate the back flanges **108** from each other in a horizontal direction, which would apply a horizontal shear load to the fasteners **18**, in addition to the vertical shear already being applied. However, the construction of the building component connector **104** inhibits this as well. The position of the force of the rigid rod **14** in relation to the location of the fasteners **18** extending through the back flanges **108** causes the back flanges to move apart. Additionally, the force of the rigid rod **14** against the interior of the cylindrical wall **116** of the rigid rod connector **106** because of the moment couple, also acts to force the back flanges apart. However, referring to FIGS. 2 and 4, the washer **150** is configured to be disposed between the rigid rod connector **106** of the connector **102** and the nut **16** on the rigid rod **14** that secures the rigid rod to the girder tie **100**. The washer **150** serves two functions. First, the washer **150** generally distributes the load applied by the nut **16** over the rigid rod connector **106**, like a conventional washer. Second, the washer **150** captures the ribs **112** to inhibit or prevent the first and second back flanges **108** from moving horizontally apart from each other when the truss T experiences the uplift forces, as described in more detail below.

(21) As shown in FIG. 4, the washer **150** includes an upper flange **152**, a lower flange **154** and a connecting element **156**. The connecting element **156** interconnects the upper and lower flanges **152**, **154**. The upper flange **152**, lower flange **154** and connecting element **156** are all generally planar. The upper and lower flanges **152**, **154** are generally parallel with one another and extend rearward from upper and lower edge margins, respectively, of the connecting element **156** to free ends thereof. The upper and lower flanges **152**, **154** are generally perpendicular to the connecting element **156**. The upper and lower flanges **152**, **154** each define an aperture or opening **158** sized and shaped to receive the rigid rod **14** through the flange. The openings **158** are aligned (e.g., vertically aligned) with one another and are configured to align with the central passage **114** of the rigid rod connector **106** when the washer is positioned on the rigid rod connector **106**. Accordingly, the upper flange **152** is disposed between the nut **16** and the rigid rod connector **106** when the washer **150** and connector **102** are attached to the rigid rod **14**.

(22) The washer **150** includes a back flange brace, generally indicated at **160**, configured to brace the ribs **112** to inhibit the first and second back flanges **108** from moving relative to one another when the truss T experiences the uplift forces. Specifically, the back flange brace **160** inhibits the first and second back flanges **108** from rotating relative to one another, as explained in more detail below. In the illustrated embodiment, the washer includes two back flange braces **160** (e.g., upper and lower back flange braces). Each back flange brace **160** is configured to engage the ribs **112** to prevent the ribs, and therefore the back flanges **108**, from moving apart from one another.

Specifically, each backflange brace **160** is configured to inhibit the ribs **112** from moving away from one another (e.g., inhibit the back flanges **108** from moving away from one another). In the illustrated embodiment, the upper and lower flanges **152**, **154** each define one back flange brace **160**. Each back flange brace **160** includes an open ended slot **162** (e.g., a slot extending from an edge margin of the upper or lower flange **152**, **154**). The slots **162** are sized and shaped to receive the ribs **112** therein. Accordingly, the slots **162** are generally aligned (e.g., vertically aligned) with one another. Opposite sides of each slot **162** are defined by bracing tabs **164** (e.g., portions of either the upper or lower flanges **152**, **154**). Each bracing tab **164** engages a respective one of the ribs **112** when the ribs are disposed in the slot **162** to prevent the back flanges **108** from moving horizontally apart.

(23) In the illustrated embodiment, each bracing tab **164** on the lower back flange brace **160** includes a projection **166** sized and shaped to mate with one of the notches **118** of a corresponding back flange **108**. The projections **166** extend rearward from a rear edge margin of the lower flange **154**. The mating of the projections **166** with the notches **118** facilitates the positioning of the washer **150** relative to the connector **102**, and further prevents the back flanges **108** from moving (e.g., toward or away) relative to one another and helps hold the washer in place relative to the connector when the girder tie **100** is subject to the uplift forces.

(24) In operation, the washer **150** is placed on the connector **102** such that the openings **158** of the washer are aligned with the central passage **114** of the connector. In this position, the upper flange **152** of the washer **150** overlies the upper end of the rigid rod connector **106** and the lower flange **154** lies under the lower end of the rigid rod connector. Accordingly, the distance between the upper and lower flanges **152**, **154** is generally the same as the height of the rigid rod connector **106**. When the washer **150** is positioned on the connector **102**, the projections **166** are inserted into the corresponding notches **118**. This facilitates the alignment of the openings **158** and central passage **114**. Moreover, when the washer **150** is positioned on the connector **102**, the ribs **112** are captured in the slots **162**. The washer **150** may be placed on the connector **102** before or after the connector is secured to the truss T with the fasteners **18**. Once positioned, the rigid rod **14** can be inserted into and extend through the openings **158** and central passage **114**. The nut **16** is then threaded onto the end of the rigid rod **14** until the nut engages the washer **150**, thereby securing the girder tie **100** to the rigid rod. Preferably, the girder tie **100** is secured to the truss T before the nut **16** is tightened down against the washer **150**.

(25) When the girder tie **100** is subject to the uplift forces, the first and second back flanges **108** are urged to move relative to one another (e.g., generally away from one another). Specifically, the first and second back flanges want to generally pivot and rotate relative to one another. This movement of the back flanges **108** is caused, at least in part, by the moment couple formed between the rigid rod connector **106** and the rigid rod **14** in conjunction with the force of the rigid rod being offset from the location of the resisting force provided by the fastener **18**. Below the point of level of the engagement of the nut **16** with the washer **150** at the top of the rigid rod connector **106**, the bottoms of the back flanges **108** are urged to pivot away from each other about a separation axis perpendicular to the back flanges and passing through the center of engagement of the nut with the washer at the top of the rigid rod connector. The back flange braces **160** inhibit this movement (e.g., lateral and/or rotational movement about the longitudinal axis of the rigid rod connector **106**) of the back flanges **108**. As the back flanges **108** try to move away from one another due to the uplift forces, the bracing tabs **164** of the back flange braces **160** engage the ribs **112**, preventing the ribs and therefore the back flanges from moving apart and causing the back flanges **108** to act as a single piece of material. Moreover, because the back flange braces **160** restrict the movement (e.g., generally horizontal movement) of the back flanges **108**, any load (e.g., horizontal load) that would have been imposed on the fasteners **18** because of the movement of the back flanges is eliminated (e.g., all this horizontal load is contained and carried by the back flange braces). This eliminates placing any additional load on the fasteners **18** and generally keeps the load on the fasteners to

generally only vertical shear and withdrawal. It is understood that by subjecting the fasteners **18** to less load (e.g., the horizontal load), the fasteners can carry or withstand a large amount of vertical shear load and withdrawal load. Thus, the back flange braces **160** help strengthen the connection of the building component connector **104** to the truss T. In addition, because the back flanges **108** pivot relative to another, a portion of the back flanges above the separation axis may move toward one another as the back flanges pivot. As a result, portions of the ribs **112** are brought into engagement with and push against one another, cancelling out a portion of the load (e.g., horizontal load).

(26) The connector **102** and washer **150** are preferably made from metal. In one embodiment, the connector **102** and washer **150** are each formed as one piece (e.g., the connector and washer are each integral one-piece components) from metal blanks **200** and **300**, respectively, (FIGS. 5 and 6) that are stamped from a sheet of material (e.g., metal) and bent into shape. As a result, the first and second back flanges **108** are indirectly connected to each other by way of the rigid rod connector **106**. However, there is no direct connection between the first and second back flanges in the plane of the first and second back flanges. A direct connection could be made between the first and second back flanges **108** in their common plane, but is not necessary because of the functionality of the washer **150**. In one embodiment, the connector **102** and washer **150** are each stamped from 10 gauge steel, although other suitable materials are within the scope of the present disclosure. In other embodiments, the connector **102** and the washer **150** are made from multiple pieces joined together, such as by welding.

(27) Referring to FIG. 7, another embodiment of a girder tie according to the present disclosure is generally indicated by reference numeral **400**. Girder tie **400** is generally analogous to girder tie **100** and, thus, for ease of comprehension, where similar, analogous or identical parts are used, reference numerals “300” units higher are employed. Accordingly, unless clearly stated or indicated otherwise, the above descriptions regarding girder tie **100** also apply to girder tie **400**.

(28) The girder tie **400** is a first or left oriented girder tie configured to be attached to the left side of a building component, such as a truss T' or a column. In this embodiment, the right or second back flange **408** has a straight outer edge margin. As a result, the width of the right back flange **408** is reduced (compared to back flange **108**), thereby reducing the overall width of the connector **402** (e.g., building component connector **404**). The back flanges **408** of the girder tie **400** have different shapes. The straight edge of the second back flange **408** allows the connector **402** to be attached to smaller building elements, such as 2×4 piece of dimensioned lumber.

(29) Referring to FIGS. 8 and 9, another embodiment of a girder tie according to the present disclosure is generally indicated by reference numeral **500**. Girder tie **500** is generally analogous to girder tie **100** and, thus, for ease of comprehension, where similar, analogous or identical parts are used, reference numerals “400” units higher are employed. Accordingly, unless clearly stated or indicated otherwise, the above descriptions regarding girder tie **100** also apply to girder tie **500**.

(30) The girder tie **500** is a second or right oriented girder tie configured to be attached to the left side of a building component, such as a truss T' or a column. In this embodiment, the left or first back flange **508** has a straight outer edge margin. As a result, the width of the left back flange **508** is reduced (compared to back flange **108**), thereby reducing the overall width of the connector **502** (e.g., building component connector **504**). This allows the connector **502** to be attached to smaller building elements, such as 2×4 piece of dimensioned lumber. FIGS. 7 and 8 illustrate the left oriented girder tie **400** attached to the left side of the truss T' and the right oriented girder tie **500** attached to the right side of the truss T'.

(31) Modifications and variations of the disclosed embodiments are possible without departing from the scope of the invention defined in the appended claims.

(32) When introducing elements of the present invention or the embodiment(s) thereof, the articles “a”, “an”, “the” and “said” are intended to mean that there are one or more of the elements. The terms “comprising”, “including” and “having” are intended to be inclusive and mean that there may

be additional elements other than the listed elements.

(33) As various changes could be made in the above constructions, products, and methods without departing from the scope of the invention, it is intended that all matter contained in the above description and shown in the accompanying drawings shall be interpreted as illustrative and not in a limiting sense.

Claims

1. A girder tie for connecting a building component to a rigid rod to resist uplift forces applied to the building component, the girder tie comprising: a connector including a building component connector configured to attach to the building component and a rigid rod connector joined with the building component connector and configured to attach to the rigid rod, the rigid rod connector configured to form a moment couple with the rigid rod to resist the uplift forces applied to the building component when the building component experiences the uplift forces; and a first nut configured to be threaded onto the rigid rod into engagement with the rigid rod connector to secure the rigid rod to the girder tie, and a second nut configured to be threaded onto the rigid rod at a location spaced from the first nut to cooperate with the first nut and rigid rod connector to block upward movement of the building component connector on the rigid rod.
2. The girder tie of claim 1, wherein the rigid rod connector is configured to engage the rigid rod at least at two longitudinally spaced apart locations on the rigid rod when the building component experiences the uplift forces to form the moment couple with the rigid rod.
3. The girder tie of claim 2, wherein the rigid rod connector includes a generally cylindrical tube defining a passage sized and shaped to receive the rigid rod.
4. The girder tie of claim 3, further comprising a washer configured to be disposed between the rigid rod connector and the first nut.
5. The girder tie of claim 4, wherein the building component connector includes first and second back flanges free of direct connection to one another, the first and second back flanges each configured to attach to the building component.
6. The girder tie of claim 5, wherein the washer includes a back flange brace configured to inhibit the first and second back flanges from moving relative to one another when the building component experiences the uplift forces.
7. The girder tie of claim 6, wherein the back flange brace inhibits the first and second back flanges from rotating relative to one another.
8. The girder tie of claim 6, wherein the building component connector includes a first rib interconnecting the first back flange and the rigid rod connector and a second rib interconnecting the second back flange and the rigid rod connector, the back flange brace configured to engage the first and second ribs to inhibit the first and second back flanges from moving relative to one another.
9. The girder tie of claim 1, further comprising the rigid rod.
10. The girder tie of claim 9, further comprising a holdown configured to attach to the rigid rod.
11. A girder tie for connecting a building component to a rigid rod to resist uplift forces applied to the building component, the girder tie comprising: a connector including a building component connector configured to attach to the building component and a rigid rod connector joined with the building component connector and configured to attach to the rigid rod, the rigid rod connector being configured to engage the rigid rod at least at two spaced apart locations on the rigid rod to form a moment couple with the rigid rod to resist the uplift forces applied to the building component when the building component experiences the uplift forces, wherein the rigid rod connector includes a generally cylindrical tube defining a passage sized and shaped to receive the rigid rod, and wherein the building component connector includes first and second back flanges free of direct connection to one another, the first and second back flanges each being configured to

attach to the building component; a nut configured to be threaded onto the rigid rod into engagement with the rigid rod connector to secure the rigid rod to the girder tie; and a washer configured to be disposed between the rigid rod connector and the nut, the washer including a back flange brace configured to inhibit the first and second back flanges from moving relative to one another when the building component experiences the uplift forces.

12. The girder tie of claim 11, wherein the back flange brace inhibits the first and second back flanges from rotating relative to one another.

13. The girder tie of claim 11, wherein the building component connector includes a first rib interconnecting the first back flange and the rigid rod connector and a second rib interconnecting the second back flange and the rigid rod connector, the back flange brace configured to engage the first and second ribs to inhibit the first and second back flanges from moving relative to one another.
